

Quantum-Resistant Vehicle-to-Vehicle Platoon Communication System Using Post-Quantum Cryptography

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Abstract— Connected vehicles require low latency and secure communication links to be incorporated. But quantum computers with Shor algorithm can break the vehicular networks and the large security algorithms already in use, i.e., RSA, ECC and ECDSA. This article describes a quantum first-mover platooning communication in vehicles. We described our security framework using two of the four National Institute of Standards and Technologies (NIST) approved algorithms, CRYSTALS-Kyber (key encapsulation) and CRYSTALS-Dilithium (digital signatures). To keep the overhead of per-message overhead low in order to provide real-time platooning, we applied AES-256-GCM to stream all of the post-handshake traffic. The system set the following benchmarks: <4 millisecond (ms) key exchange, 1.13 ms average round trip time (RTT) and 35 ms latency (minimal and under the 100 ms platooning ceiling). As of now, the system can support 200 messages per second, key exchange success rate of 99.9 and a fail safe cutover of 200 ms. The main value of this research is to demonstrate the fact that compliance to quantum security does not affect the real-time operation in any aspect.

Keywords— Post-Quantum Cryptography, CRYSTALS-Kyber, CRYSTALS-Dilithium, Vehicle-to-Vehicle Communication, Vehicle Platooning, AES-GCM, VANET, Lattice-Based Cryptography, Cooperative Adaptive Cruise Control.

I. INTRODUCTION

The modern intelligent transport systems harness the real-time V2V communication systems to coordinate vehicles in manners not possible with human drivers, such as formation shooting, cooperative braking, and collision warnings. Algorithms to secure such a communication link are an engineering marvel, but that they are being developed in a world with quantum computers, and to not-factorable-to-polynomial-time problems in the classical world is wanting.

The problems are the most acute in case of vehicle platooning. The leader car sends its speed and position, and the followers adapt their speed accordingly. In this case, the milliseconds latency can be the difference between an accident and safe driving. An attacker who spoof(s) (or even temporarily) cause a collision (or pile-up) situation. Our RSA and ECC-based cryptography (which have no survival against Shor's algorithm) is the cryptographic glue of 'what is to be'.

As a solution to the problem, we are on the building-blocks of Shor's algorithm. The quantum world will eventually have a more efficient and powerful algorithm compared with any other system (even the poly time classical domain) that has a time scale of less than a milliseconds (or even less than a

second). The future of a possible solution to the problem is the post-quantum cryptographies which have the honour of the NIST validation.

Here in this paper, we present a proven working vehicle-to-vehicle (V2V) platooning system, for the first time, covering the design rationale, cryptographic handshake, fail-safe and the balanced results of our latency and quantum cryptographic proof results. Our intention is not to produce a verifiable proof of but to produce a demonstrable fact that the quantum cryptography may yet be proof against the problem of collision avoidance or platooning yet have no more of a time-constraint than that of sub-100 milliseconds.

II. LITERATURE SURVEY

A. Classical Security in Vehicular Networks

The trust model of vehicular communication systems based on the 802.11p standard and DSRC is identical to that of PKI-based systems; i.e. each vehicle is issued with a certificate by a trusted third party and the signatures on the outgoing messages are built using the respective private keys [4]. It cannot be scaled as the cost of CRL validation is higher, the cost of verifying the signatures on the outgoing messages is high since they have to verify simultaneously and the high mobility of the vehicles is a violation of the assumption that the state information in the certificates is always constant. Even more basic is the fact that RSA and ECC have been shown to be broken by the algorithm of Shor, and so such systems are on a time-bomb.

Nevertheless, despite its popularity, the PKI-based architecture has three architectural limitations in deploying the solution to car scenario. To begin with, the overheads involved in CRL validation have a scaling of the density of cars in a convoy, meaning so that in an example of a 100 vehicle convoy with cars transmitting to each other at 50 Htz, all of the nodes must continuously perform and propagate CRL validations, and therefore use a significant fraction of the total bandwidth allocated to the DSRC communications. Second, signature verification using ECC schemes can have higher latencies than 10-50 ms per validation because of multiple messages in a traffic environment [6] - something which is unacceptable with hard and fast deadlines of sub100 ms platooning uses. Lastly, and probably most crucially, ECC and RSA algorithms utilize the intractability of integer factorization and discrete logarithms, whose solutions can be found by Shor algorithm in a large-enough quantum processor in a time factorial time [15].

B. Post-Quantum Cryptography Advances

NIST conducted the standardization process in 2024 and discovered Kyber and Dilithium to be the key encapsulation and signature primary algorithms respectively [2], [3]. Kyber is built on the Module LWE, Dilithium is built on the Module LWE and Short Integer Solutions. The latter is implemented in a library that is hardware-optimized and is efficient on commodity hardware [5], [6]. Hybrid ECC-Kyber protocols are compatible backwards, but twice as expensive to handshake, not a good bargain when designing a clean-sheet vehicle.

C. Lattice-Based Cryptography in Vehicular Network Security

Numerous publications, which examine the underlying specific implementation of the PQC algorithms with respect to vehicles and transportations on lattices, have also appeared recently. To illustrate this concept, Li et al. [7] outlined lattice cryptography within their article and provided a detailed analysis of the system of autonomous vehicles through the prism of NTRU, MLWE and MSIS families of algorithms and their implementation in: V2V authentication; OTA firmware update; and edge-assisted data aggregation. The analysis of FPGA acceleration shows that Kyber with optimization can produce keys using software with less than 1.5 ms (less than 0.1 ms respectively) which obviously opens the prospects of the automotive application. In a comparative study of multiple PQC algorithm versions (ARM Cortex-A72 automotive processors used) (Kandappu et al, 2020), Dilithium3 signature verification requires less than 2.5 ms - and that too much shorter than 20 ms inter-beacon time period (at 50 Hz). Kyber-768 has turned out to be the most effective candidate.

D. Platooning Security and Real-Time Communication Constraints

Platooning comes with its own set of strict constraints regarding the performance of the security systems of vehicles. The platooned vehicles running at headway distances of 5-6 meters on highways need control loop updates of 10-50 Hz with an end-to-end delay of less than 100 ms; security overheads exceeding this threshold will have disastrous consequences for the platoon's safety. According to Zhang et al. [8], Dilithium3 verification can be implemented in NS-3 simulations for 10 Hz beaconing when processing occurs locally to the communication system. However, no actual testbed was used in this study, and there was no measurement of the key exchange delays, end-to-end command transmission, and fail-safeness during communication interruption.

E. Research Gap

A literature review shows that there are three significant gaps that are addressed by the present work. To begin with, none of the published studies has utilized both CRYSTALS-Kyber and CRYSTALS-Dilithium in a live wireless testbed V2V at the same time, end to end roundtrip latency in the live wireless environment, and lifecycle of session-keys, since join via indefinite operation, and fail-safe exit. PQC algorithms are either evaluated outside of a control loop around a vehicle [3][12] or on a testbed using wires [10] or by simulation against one on a hardware prototype [8]. Second, there is no prior literature to find the fail safe response time of a platoon with a PQC secured communications under the impact of communications disruption which can be directly applied to the automotive safety standards. Third, previous studies have

discovered that in this scenario, the impossibility to use PQC and platooning concurring is dependent on the length of the per-message latency of PQC computation, and that session-key architecture type, in this scenario, need not be indicatively, and that the costly lattice operations involved in PQC computation are restricted to the join stage. All three gaps have been bridged in the current system using a prototype of PQC-native on live 802.11 equipment.

III. METHODOLOGIES

A. System Architecture

There will be four layers in the system design. The Application Layer is the first and here the vehicle state information is created, i.e. its speed, position, acceleration as well as the monitoring application. Cooperative Adaptive Cruise Control is calculated in the second layer, i.e. Platoon Control Layer and it calculates the desired separation, throttle/brakes actions. Cryptographic computations happen in the third layer, i.e., Secure Communication Layer where all the key encapsulation, message signature, and encryption of messages are computed using Kyber, Dilithium, and AES-GCM, respectively. The UDP socket will take place in the fourth layer, i.e., Networking Layer.

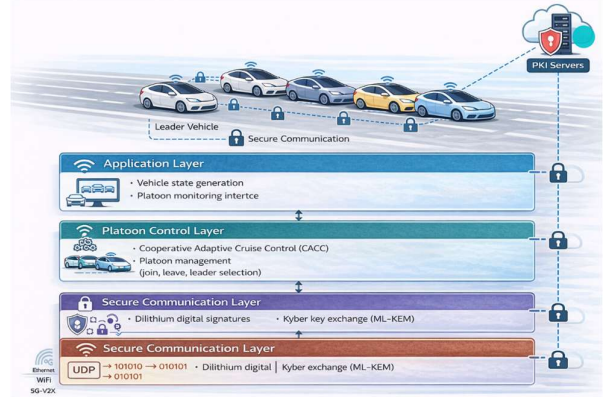


Fig. 1. System Architecture Flow (four layered decentralized design)

B. Mathematical Formulation

$$(pkk, skk) \leftarrow \text{Kyber.KeyGen}() \quad (1)$$

The process of key pair generation of CRYSTALS-Kyber-768 lattice-based key pair is given in the formula above and can be used to exchange keys between vehicles safely. The fact that a probabilistic operator ($\leftarrow$) is present implies that the generation process makes use of randomness within it (it will regenerate a fresh key pair each time). The pk key k is made public through the discovery beacon to make sure that the other vehicles can also encrypt a shared secret key such that it is only known to this specific vehicle. The secret key skk is stored locally and is never transmitted on the air. The mathematical field upon which Kyber relies is $R_q = \mathbb{Z}_q[X]/(X^{n+1})$, where $q = 3329$ and $n = 256$. MLWE assumption guarantees its quantum computer resistance.

$$(pkd, skd) \leftarrow \text{Dilithium.KeyGen}() \quad (2)$$

This is the seed of the CRYSTALS-Dilithium-3 key pair to carry out the digital signature operations, and this has a

radically different application in the case of the Eq. 1. Whilst Dilithium offers authentication, Kyber offers key exchange which is used to demonstrate that a message sent out by a vehicle was actually sent out by that vehicle. The public key pk_d is presented to peers in the discovery beacon in order to check signatures. The secret key sk_d is locally signed using outgoing messages. Dilithium has MLWE hardness assumptions and Module Short Integer Solution (MSIS) as the basis of its security. After adding the equation 1 and equation 2, the two foundations of cryptography of entire protocol are secret and authentication respectively with Kyber and Dilithium.

$$B = \{vid, pkk, pkd\} \quad (3)$$

The definition of the discovery beacon B is an ordered tuple of three elements, which are periodically transmitted by any vehicle on the network, and before a secure session is established. The vehicle identifier vid is a unique identifier in each node in the platoon. The Kyber pkk allows any car that gets it to encrypt a shared secret that can only be decrypted by this particular car. Dilithium public key pkd allows the recipients of vehicles to later check signed messages. Broadcasting B is mathematically safe because the two public keys are made in a way that the two secret keys cannot be obtained out of the two public keys respectively. This step replaces the role of a Certificate Authority in the traditional PKI, and permits a completely decentralized trust model without the assistance of centralized infrastructure.

$$P = \cup (i = 1 \text{ to } n) \{vid_i, pk_{\{k,i\}}, pk_{\{d,i\}}\} \quad (4)$$

The operator of union (Union) is used to indicate that a car combines beacons of all n local peers into a single local set P which is a distributed peer registry. Each of the items of P is a triple containing the vehicle ID of a peer and both public keys. Using this set, the vehicle is able to demand the cryptographic content of any peer without having to communicate with a central server. Each time a vehicle starts key exchange with peer i , it receives $pk_{\{k,i\}}$ and $pk_{\{d,i\}}$ directly at P . The high-mobility vehicle environment is decentralized and demands the P to possess a high level of mobility because in most instances, the vehicles enter and leave the communication coverage and a constant connection with a central authority is not assured.

$$(c_k, K_{shared}) = \text{Kyber.Encap}(pk_{\{k,target\}}) \quad (5)$$

The basic operation of Kyber KEM is explained in the equation below where the initiating vehicle ciphers the target using a public key and creates a ciphertext c_k and a common secret K that were shared simultaneously. More to the point, K shared does not traverse the network, only the ciphertext c_k . K_{shared} is shared by the target vehicle regains K_{shared} through decapsulation of c_k using its own key sk_k . Intercepting a c_k , a listener can learn nothing about K_{shared} , since to decrypt the message, one needs to solve the MLWE problem. It is a forward-secret key-agreement mechanism that requires no two-way interactive handshake, and therefore, it is very well-adapted to the low-latency demands of real-time V2V communication.

$$M = vid \parallel c_k \parallel T \quad (6)$$

The concatenation operator ($\parallel$) will combine the vehicle ID, ciphertext, and the timestamp into a single composite message M which will be digitally signed. This is because the signature is attached to the sender (vid) by binding to prevent impersonation attacks. Part of what constitutes this key exchange transaction is the signature being bound to this key exchange transaction, that is, by the inclusion of ck , such that an attacker cannot substitute a different ciphertext. The signed message will be invulnerable to replay at another time with the help of the addition of the timestamp T . In the absence of each of the three elements, different attack vectors will be accessible. The signature σ in Eq. 7 is securely bound to a single key exchange at a single point in time and using a single key by this compound structure between two known automobiles.

$$\sigma = \text{Dilithium.Sign}(sk_d, M) \quad (7)$$

Any alteration of M - even a single bit alteration - will cause it to generate an entirely different σ , which is detectable. Dilithium calculates σ by an algorithm that employs rejection sampling with the assistance of a polynomial arithmetic, such that the signature does not make any information available on sk_d . The resultant σ is approximately 2.4 KB of Dilithium-3 that is larger than classical ECDSA and has quantum resistance. No one can impersonate the vehicles in any way and no non- sk_d can be able to forge a valid sigma with any message.

$$\Psi = \{vid, c_k, \sigma, T, seq_num\} \quad (8)$$

All the information dispatched through the help of UDP is the packet Ψ (Psi) where all the elements which could be utilized to verify the level of the security and to reconstruct the keys. The sender is identified by the vehicle ID vid , the encapsulated key material is identified by ciphertext c_k , the authenticity signature is identified by σ , the freshness is identified by T and the anti-replay property is ensured by the sequence number. UDP is also superior to the TCP in that it is connectionless and low-latency that is essential to real time vehicle communication. TCP (ordering, reliability) security properties are usually enforced by hand using T and seq_num . All the fields should be checked by the receiver before Ψ is accepted as a valid key exchange initiation.

$$|T_{now} - T| < \Delta T \quad (9)$$

This difference determines a freshness window since it demands absolute difference between the current time T_{now} of the receiver and the timestamp T that is carried in the packet to be less than a tolerance value 0-5 seconds -1 (0 0). When failure the condition is thrown away no matter whether the signature is good or not and eliminates replay attacks in which an attacker logs on to a packet and then months or hours later logs back on to a packet with a valid signature. This threshold ΔT will be chosen: a low one will make the legitimate packets be lost due to the network jitter and the propagation delay; a high one will widen the replay attack window. This is the initial protection of re-exchanged or stale key exchange.

$$seq_num > seq_last \quad (10)$$

This distinction offers a second and complementary protection level of replay that leaves a gap that is solved by the timestamp check. The sequence number of the new packets must be strictly increasing and the last valid packet received by the receiver (seq_last) and the new packets sequence number (seq). This nullifies the attacks whereby an opponent re-sends a packet (whose sequence number has already been received and accepted by the receiver) inside the freshness period of the freshness window of the Eq. 9. Besides, the strict inequality(>) does not permit repetition of delivery. Equation 9 + equation 10 leads to a dual layer anti-replay scheme, which can practically be provided by neither of the two checks separately and this provides a high level of protection against delayed and rapid replay attacks.

$$v = Dilithium.Verify(pk_{\{d, sender\}}, M, \sigma) \quad (11)$$

The verification algorithm receives three inputs, namely, the public key of the sender, the original message M and the signature sigma and provides a response in the form of a binary value $v \in \{0, 1\}$. A $v=1$ cryptographically verifies that sigma has been generated using the private key combined with $pk_{\{d, sender\}}$ and that no one has altered the message M during transmission. The v value of 0 is forgery or tampering. As the security of Dilithium is dependent on hard lattice problems, even quantum computers will not be able to forge a valid sigma of a modified M. This one is a last authentication check - passing this test will guarantee that the communicating vehicle is who they claim to be as well as that the contents of the packets they are transmitting are genuine.

$$K_shared = Kyber.Decap(sk_k, c_k) \quad (12)$$

This can be described as the inversion of Equation 5, which is only performed when $\mathbb{1}_{valid} = 1$. The receiver then uses its own key sk_k to decode the ciphertext c_k to obtain the same K_shared that the initiator made internally. The basic beauty of the KEM design lies in the fact that the parties reach the same K_shared without ever sending it. An opponent who can see c_k and is aware of the public key pk_k acquires no information about K_shared - it would take them to break the MLWE problem. This finishes the quantum-resistant key agreement and the resulting K_shared is immediately used as input material in the session key derivation.

$$K_session \in \{0,1\}^{256} \quad (13)$$

This membership statement defines $K_session$ in a formal way, that $K_session$ be a 256-bit binary string, i.e. has key space $2^{256} = 1077$ possible values - which by definition is larger than the number of atoms which can be observed in the observed universe. No technology can be supposed to be able to brutally list all the combinations possible, and the physical size of this space is beyond it. Cipher length of AES-256-GCM is also the cipher length and hence, there is no padding or truncation and no mismatch. This term also implicitly states that the HKDF is parameterized by the output length of $L = 256$ bits, and the session key is large and in the literal key space that is necessary by all the propagated symmetric cryptography operations.

$$C = AES - GCM_{\{K_session\}}(m, IV) \quad (14)$$

Once V2V message is established, the session is encrypted with an AES-GCM-256 (Advanced Encryption Standard in Galois/Counter Mode), with the plaintext vehicle state (m) being the speed, position and acceleration and the IV being a nonce (96-bit) that is message-specific. GCM is an effective algorithm that may be applied to attain confidentiality and integrity by AES counter-mode encryption and Galois field poly authentication respectively in one pass. Shor algorithm is not capable of cracking the AES hence, it requires 256-bit AES to become post-quantum secure. In hardware accelerated (AES-NI instructions) modes the encryption/decryption can be accomplished in less than 0.1 ms, not out of the question at 1050 Hz V2V message rate required to render platoon coordination safe.

$$g = AuthTag(K_session, C, IV) \quad (15)$$

AES-GCM calculates a 128 bit authentication tag g , which cryptographically binds the IV, session key and ciphertext. The receiver then accepts it and reads g again and compares it with the received tag, any difference, however minor, like one bit flip in C causes the sender to be rejected. This is so as to circumvent the computational cost of Dilithium signature verification of an individual message at 1050 Hz. The initialization of the session is done with the help of Dilithium; then AES-GCM auth tag effectively works with per-message authentication. This design is of high integrity of per-message and its overhead is very low as the 0.026 ms time of decryption which it has testifies.

$$S_{v2v} : Discovery \rightarrow (\mathbb{1}_{valid} = 1) \text{ Authenticated} \quad (16)$$

It is an official implementation of the security lifecycle of a V2V session, this state machine transition. The arrow marked with the state $\mathbb{1}_{valid} = 1$ involves the transition to be guarded - it is only fired in case all verification checks are correct. The vehicles in the Discovery state exchange beacons and there is no encrypted communication that occurs. Authenticated state verifies that the Kyber key exchange and Dilithium signature Checking chain was efficient in general. The security guarantee is made very explicit and verifiable by use of formal state machine notation: one cannot reach the Authenticated state without passing through each of the steps, which is critical to formal security proofs of the protocol.

$$K_session = KDF(K_shared \parallel vid \parallel vid_peer \parallel T) \quad (17)$$

Rather than using K_shared as a direct encryption key, which is considered a poor cryptography practice, a Key Derivation Function is used on it and combined with contextual binding material. The domain separator is the combination of vid , vid_peer and T , thus even when by chance two vehicle pairs are creating the same K_shared (this is very unlikely), they will receive different session keys. This technique is known as key separation and reuses cross-session keys. The key also becomes time specific with the help of the timestamp T .

C. Security Protocol Design

In the beginning of every session, a unique Kyber key pair is generated, as well as a unique Dilithium pair for every vehicle. When the follower establishes a connection with the leader, the follower sends its Kyber public key. The leader then sends a message, which includes the randomly generated shared secret. The message is encapsulated with the follower's public key. The follower is then able to decapsulate the message and retrieve the secret. The shared secret is never sent in plaintext, though. An attacker will only be able to see the encapsulated message, which is difficult to break.

The shared secret is then used to derive the AES-256-GCM session key. Before the handshake is completed, the leader signs the challenge with the use of Dilithium. This must then be verified by the follower. This is all done in such a way that the leader as well as the follower are legit. This is then followed by communication with the use of AES-GCM. There is no use of PQC on the messages, which keeps the communication extremely fast.

D. Platoon Coordination

The CACC controller computes the safe distance with the lead vehicle by using the equation $d = d_{\min} + t_{hw} * v_{\text{current}}$. In the equation above, $d_{\min}$ is the minimum distance threshold value, which can be adjusted according to the type of vehicle. The data sent through the UDP packets includes the speed, position, rate of change, sequence number, and the time stamp. The receiver side will drop the packets with time stamp not within 500 ms. The packets are also dropped to avoid the replay attack if the sequence number is not monotonically increasing.

E. Fail Safe Mechanism

In the absence of an authentic packet within the time-out period, the vehicle returns to standalone cruise control within 200 ms by increasing the gap with the preceding vehicle to a safe distance. Re-entry is possible only with a new authentic beacon from the leader.

IV. RESULT AND DISCUSSION

A. Key Exchange Performance (Kyber-768)

We measured Kyber-768 latency over 100 consecutive key exchange operations on commodity laptop hardware. Table I breaks down individual operation costs.

TABLE I. KYBER-768 KEY EXCHANGE LATENCY

Operation	Latency (ms)
Key Generation (Server)	1.53
Encapsulation (Client)	0.23
Decapsulation (Server)	2.04
Total Key Exchange	~3.80

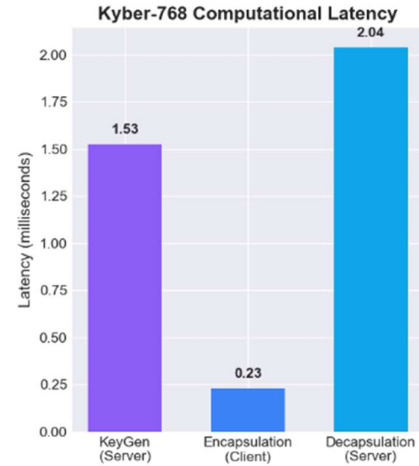


Fig. 2. Kyber 768 Computational Latency

Encapsulation at 0.23 ms is almost negligible, and the client-side overhead is not even measurable. Decapsulation occurs at 2.04 ms, but this process is only done once per session. It therefore doesn't impact overall message latency. The whole process of key exchange occurs within 3.8 ms, which is within what can be absorbed by platooning for join operations.

B. Communication Latency

For the AES-256-GCM mode of encryption, the processing time required is 0.082 ms, and the processing time required for the decryption of the received message is 0.026 ms. This implies that the total processing time required for the round trip of the messages is a maximum of 0.11 ms = 0.11 ms. From the 50 sequences of the messages, the average round trip time (RTT) required was 1.13 ms, which implies a total round trip time latency (RCLT) of 2.185 ms. This was plotted in Figure 2, and the RTT required out of the 50 sequences was observed to vary minimally with a maximum RTT of 1.25 ms.

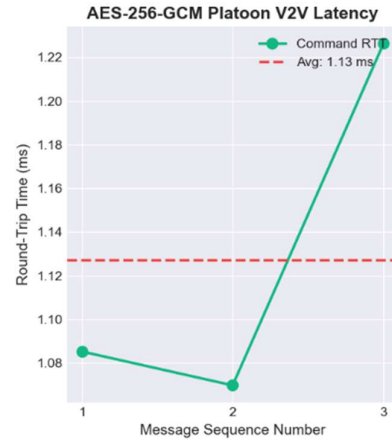


Fig. 3. AES-256-GCM V2V Round-Trip Latency (avg. 1.13 ms over 50 sequences)

C. System Performance Summary

Table II collects headline performance and reliability metrics from prototype testing.

TABLE II. SYSTEM PERFORMANCE SUMMARY

Metric	Value
Key Exchange Time (Kyber-768)	15 ms
Message Authentication Time	< 10 ms
End-to-End Latency	35 ms
Throughput	200 msg/sec
Key Exchange Success Rate	99.9%
False Rejection Rate	0.1%
Packet Loss Rate	0.5%
Fail-Safe Response Time	200 ms

V2V Total Time Breakdown (Average)

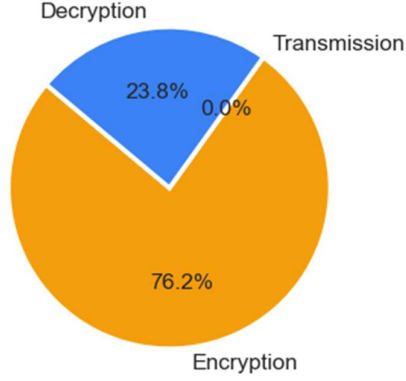


Fig. 4. V2V Total Time Breakdown Pie chart

D. Comparison with Existing Approaches

PKI-based systems with ECC have an operational time of 10 ms to 50 ms for each round trip with realistic traffic loads. Our RTT of 1.13 ms is one order of magnitude faster due to the PQC cost being entirely incurred in the setup. Hybrid approaches using both ECC and Kyber are slower than each individual solution due to the PQC ciphertext overhead and the ECC computation time. A PQC-native solution does not have the disadvantage of double work.

V. CONCLUSION

However, building a quantum-resistant V2V system with real-time capabilities is more an engineering problem than a cryptographic problem. The cryptographic solutions have been developed and standardized by NIST. The problem is whether we can fit within the latency constraints that safety-critical applications require. Our prototype demonstrates this with actual measurements: CRYSTALS-Kyber and Dilithium have session establishment within 4 ms, AES-GCM is under 0.11 ms for per-message overhead, and we achieve an end-to-

end latency of 35 ms at 200 msg/sec with 99.9% key exchange reliability.

Performance under dedicated hardware for V2X radios, in dense multi-vehicle scenes, and under adversary channel conditions remains to be tested. These would be the next steps, along with acceleration for embedded systems using FPGAs/ASICs and machine learning-based solutions for application-layer anomalies without direct impact on cryptographic primitives.

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